Rough Path Theory

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July 17, 2026

Preface

This is the reading note of the lecture notes *Rough Path Theory* by Andrew L. Allan, which covers the first eight sessions of the note. The note is edited by the author with the support of ChatGPT. The material in Section 2.3, and Section 2.4 is also supplemented by the textbook *A Course on Rough Paths with an Introduction to Regularity Structures*. Topics of the note include: Rough paths and rough integrals; properties of rough integration; rough differential equations; and connection with stochastic calculus.

Contents

Chapter 1 Rough Paths and Rough Integrals

1.1 Definitions and Basic Properties

Throughout these notes, V, W, and U denote Banach spaces. Whenever tensor products occur, we fix a completed tensor product equipped with an admissible crossnorm, so that $|v \otimes w| \leq |v|\,|w|$. We identify continuous bilinear maps on $V \times V$ with continuous linear maps on $V \otimes V$ whenever this causes no ambiguity.

For a path $X\colon [0,T] \to V$ and $0 \leq s \leq t \leq T$, write

$$X_{s,t} := X_t - X_s.$$

Let $\mathcal{C}([0,T];V)$ denote the space of continuous V-valued paths, and write $\|X\|_\infty := \sup_{t\in[0,T]} |X_t|$.

Definition 1.1.1 (Hölder paths). For $\alpha \in (0,1]$, define

$$\|X\|_\alpha := \sup_{0\leq s<t\leq T} \frac{|X_{s,t}|}{|t-s|^\alpha}.$$

The space of paths for which this seminorm is finite is denoted by $\mathcal{C}^\alpha([0,T];V)$.

Remark. The map $X \mapsto |X_0| + \|X\|_\alpha$ is a norm under which $\mathcal{C}^\alpha([0,T];V)$ is complete. The seminorm alone does not detect additive constants.

If $0 < \alpha < \beta \leq 1$, the inclusion $\mathcal{C}^\beta([0,T];V) \hookrightarrow \mathcal{C}^\alpha([0,T];V)$ is continuous. It is strict whenever $T > 0$ and V is nontrivial. The Hölder spaces are generally nonseparable in their natural Hölder norms.

Lemma 1.1.2 (Lower semicontinuity). Let $X^n \in \mathcal{C}^\alpha([0,T];V)$ and suppose that $X^n_t \to X_t$ for every $t \in [0,T]$. Then

$$\|X\|_\alpha \leq \liminf_{n\to\infty} \|X^n\|_\alpha.$$

Proof. For each fixed $s < t$,

$$\frac{|X_{s,t}|}{|t-s|^\alpha} = \lim_{n\to\infty} \frac{|X^n_{s,t}|}{|t-s|^\alpha} \leq \liminf_{n\to\infty} \|X^n\|_\alpha.$$

Taking the supremum over $s < t$ gives the result. □

Lemma 1.1.3 (Interpolation). If $0 < \alpha < \beta \le 1$, then every path X satisfies

$$\|X\|_\alpha \le \|X\|_\beta^{\alpha/\beta} \left(\sup_{0 \le s < t \le T} |X_{s,t}| \right)^{1-\alpha/\beta}.$$

Proof. For each $s < t$,

$$\frac{|X_{s,t}|}{|t-s|^\alpha} = \left(\frac{|X_{s,t}|}{|t-s|^\beta} \right)^{\alpha/\beta} |X_{s,t}|^{1-\alpha/\beta}.$$

Take the supremum. □

Lemma 1.1.4. Let $0 < \alpha < \beta \le 1$. Suppose $X^n \in \mathcal{C}^\beta$, $\sup_n \|X^n\|_\beta < \infty$, and $X^n \to X$ uniformly. Then $X \in \mathcal{C}^\beta$ and $\|X^n - X\|_\alpha \to 0$.

Proof. Lower semicontinuity gives $X \in \mathcal{C}^\beta$ and hence $\sup_n \|X^n - X\|_\beta < \infty$. Apply the interpolation inequality to $X^n - X$ and use $\sup_{s<t} |(X^n - X)_{s,t}| \le 2\|X^n - X\|_\infty \to 0$. □

Lemma 1.1.5 (Compact embedding). Let $0 < \alpha < \beta \le 1$ and assume that V is finite dimensional. If $X^n \in \mathcal{C}^\beta([0,T];V)$ and

$$\sup_n \left(|X_0^n| + \|X^n\|_\beta \right) < \infty,$$

then some subsequence converges in $\mathcal{C}^\alpha$ to a path in $\mathcal{C}^\beta$.

More generally, the same conclusion holds in an arbitrary Banach space if the set $\{X_t^n : n \ge 1\}$ is relatively compact in V for every $t \in [0,T]$.

Proof. The assumptions imply uniform boundedness and equicontinuity. In finite dimensions, bounded point sets are relatively compact; in the stated Banach space generalization this is assumed. The Banach-valued Arzelà–Ascoli theorem therefore yields a uniformly convergent subsequence. The preceding lemma then upgrades uniform convergence to convergence in $\mathcal{C}^\alpha$. □

Remark. Thus $\mathcal{C}^\beta([0,T];V)$ embeds compactly into $\mathcal{C}^\alpha([0,T];V)$ when V is finite dimensional. This assertion is false for a general infinite-dimensional Banach space without pointwise relative compactness.

Set

$$\Delta_{[0,T]} := \{(s,t) \in [0,T]^2 : s \le t\}.$$

For a Banach space E, let $\mathcal{C}_2(\Delta_{[0,T]}; E)$ denote the space of continuous two-parameter maps, and define

$$\|A\|_\gamma := \sup_{0 \le s < t \le T} \frac{|A_{s,t}|}{|t-s|^\gamma}.$$

The corresponding Hölder space is denoted by $\mathcal{C}_2^\gamma$. For a one-parameter path, $X_{s,t}$ denotes an increment; for a two-parameter map, $A_{s,t}$ denotes evaluation at (s,t).

Definition 1.1.6 (Hölder rough path). Let $\alpha \in (1/3, 1/2]$. An α-Hölder rough path over V is a pair $\mathbf{X} = (X, \mathbb{X})$ with

$$X \in \mathcal{C}^\alpha([0,T]; V), \qquad \mathbb{X} \in \mathcal{C}_2^{2\alpha}(\Delta_{[0,T]}; V \otimes V),$$

satisfying Chen's relation

$$\mathbb{X}_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$$

for every $0 \le s \le u \le t \le T$. The space of such rough paths is denoted by $\mathscr{C}^\alpha([0,T]; V)$.

The second level $\mathbb{X}_{s,t}$ should be viewed as a prescribed value for $\int_s^t X_{s,r} \otimes \mathrm{d}X_r$. If X is smooth, its canonical lift is

$$\mathbb{X}_{s,t} := \int_s^t X_{s,r} \otimes \mathrm{d}X_r,$$

where the integral is Riemann–Stieltjes; Chen's relation follows by splitting the integral at an intermediate time.

Definition 1.1.7 (Rough-path distance). For $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (\widetilde{X}, \widetilde{\mathbb{X}})$, define

$$\|\mathbf{X}; \widetilde{\mathbf{X}}\|_\alpha := \|X - \widetilde{X}\|_\alpha + \|\mathbb{X} - \widetilde{\mathbb{X}}\|_{2\alpha}, \qquad \|\mathbf{X}\|_\alpha := \|X\|_\alpha + \|\mathbb{X}\|_{2\alpha}.$$

The first expression is a pseudometric because it ignores the initial point of X. Adding $|X_0 - \widetilde{X}_0|$ gives a metric. With the norm

$$|X_0| + \|X\|_\alpha + \|\mathbb{X}\|_{2\alpha},$$

$\mathcal{C}^\alpha \times \mathcal{C}_2^{2\alpha}$ is Banach. Chen's relation is closed under convergence, so $\mathscr{C}^\alpha$ is a complete metric space. It is not a vector space because Chen's relation is nonlinear.

If $F\colon [0,T] \to V \otimes V$ is 2α-Hölder and $F_{s,t} := F_t - F_s$, then $(X, \mathbb{X} + F_{\cdot,\cdot})$ is again a rough path. Conversely, if two rough paths have the same first level, the difference of their second levels is additive and therefore is the increment of a 2α-Hölder path. Thus an enhancement is determined only up to such an additive perturbation.

Lemma 1.1.8 (Interpolation for rough paths). Let $1/3 < \alpha < \beta \le 1/2$. Suppose $\mathbf{X}^n = (X^n, \mathbb{X}^n) \in \mathscr{C}^\beta$ and

$$\sup_n \big(\|X^n\|_\beta + \|\mathbb{X}^n\|_{2\beta}\big) < \infty.$$

If $X^n_{s,t} \to X_{s,t}$ and $\mathbb{X}^n_{s,t} \to \mathbb{X}_{s,t}$ pointwise on $\Delta_{[0,T]}$, then $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\beta$ and $\|\mathbf{X}^n; \mathbf{X}\|_\alpha \to 0$.

Proof. Passing to the pointwise limit in the Hölder bounds and in Chen's relation shows that $\mathbf{X} \in \mathscr{C}^\beta$. The uniform Hölder bounds make the families $(s,t) \mapsto X^n_{s,t}$ and $(s,t) \mapsto \mathbb{X}^n_{s,t}$ equicontinuous on the compact simplex. For the second level, this follows by changing one endpoint at a time and applying Chen's relation together with the uniform first- and second-level bounds. Pointwise convergence of an equicontinuous family on a compact set is therefore uniform. If

$$\varepsilon_n := \sup_{s<t} \big(|X^n_{s,t} - X_{s,t}| + |\mathbb{X}^n_{s,t} - \mathbb{X}_{s,t}| \big),$$

then $\varepsilon_n \to 0$. Interpolation gives

$$|X^n_{s,t} - X_{s,t}| \lesssim \varepsilon_n^{1-\alpha/\beta} |t-s|^{\alpha},$$

$$|\mathbb{X}^n_{s,t} - \mathbb{X}_{s,t}| \lesssim \varepsilon_n^{1-\alpha/\beta} |t-s|^{2\alpha},$$

which proves convergence in the α-Hölder rough-path metric. □

1.2 Weakly Geometric Rough Paths

For the canonical lift of a smooth path $X = (X^1, \dots, X^d)$, integration by parts yields

$$\mathbb{X}^{ij}_{s,t} + \mathbb{X}^{ji}_{s,t} = X^i_{s,t} X^j_{s,t}.$$

Equivalently,

$$\mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}.$$

This identity is the second-level expression of first-order calculus.

Definition 1.2.1 (Weakly geometric rough path). A rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^{\alpha}$ is weakly geometric if

$$\mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}$$

for every $(s,t) \in \Delta_{[0,T]}$. The corresponding space is denoted by $\mathscr{C}^{\alpha}_g$.

Definition 1.2.2 (Geometric rough path). The space $\mathscr{C}^{0,\alpha}_g$ of geometric α-Hölder rough paths is the closure, in the α-Hölder rough-path metric, of canonical lifts of smooth paths.

Remark. When V is finite-dimensional,

$$\mathscr{C}^{\beta}_g \subset \mathscr{C}^{0,\alpha}_g \subset \mathscr{C}^{\alpha}_g \subset \mathscr{C}^{\alpha}, \qquad 1/3 < \alpha < \beta \leq 1/2.$$

The inclusions need not be equalities. In a general Banach space, approximation by smooth canonical lifts may require additional tensor-product and approximation assumptions, so no blanket inclusion is asserted here.

1.3 Young Integration

Theorem 1.3.1 (Sewing lemma). Let E be a Banach space and let $A\colon \Delta_{[0,T]} \to E$ be continuous. Put

$$\delta A_{s,u,t} := A_{s,t} - A_{s,u} - A_{u,t}.$$

Suppose that for some $\varepsilon > 0$ and $\lambda \geq 0$,

$$|\delta A_{s,u,t}| \leq \lambda |t-s|^{1+\varepsilon}$$

for all $s \leq u \leq t$. Then there is a unique continuous additive map $\mathcal{I}A\colon \Delta_{[0,T]} \to E$ such that

$$|(\mathcal{I}A)_{s,t} - A_{s,t}| \leq C_\varepsilon \lambda |t-s|^{1+\varepsilon}.$$

Moreover, for every sequence of partitions π of $[s,t]$ with $|\pi| \to 0$,

$$(\mathcal{I}A)_{s,t} = \lim_{|\pi|\to 0} \sum_{[u,v]\in\pi} A_{u,v}.$$

Consequently, there is a continuous path γ, unique after imposing $\gamma_0 = 0$, such that $\gamma_t - \gamma_s = (\mathcal{I}A)_{s,t}$.

Proof. On $[s,t]$, let $A^n_{s,t}$ be the sum of A over the dyadic partition with 2^n intervals. Splitting every interval at its midpoint gives

$$|A^{n+1}_{s,t} - A^n_{s,t}| \leq \lambda |t-s|^{1+\varepsilon} 2^{-n\varepsilon}.$$

Hence $A^n_{s,t}$ converges and

$$|(\mathcal{I}A)_{s,t} - A_{s,t}| \leq \frac{\lambda}{1-2^{-\varepsilon}} |t-s|^{1+\varepsilon}.$$

The same estimate on each interval of an arbitrary partition π gives

$$\left| (\mathcal{I}A)_{s,t} - \sum_{[u,v]\in\pi} A_{u,v} \right| \leq C_\varepsilon \lambda \sum_{[u,v]\in\pi} |v-u|^{1+\varepsilon} \leq C_\varepsilon \lambda |t-s|\, |\pi|^\varepsilon.$$

Thus arbitrary Riemann sums converge to the same limit. Concatenating partitions of $[s,u]$ and $[u,t]$ proves additivity. Continuity follows from the estimate, and uniqueness follows because the difference of two such additive maps is $O(|t-s|^{1+\varepsilon})$; summing over finer partitions forces that difference to vanish. $\square$

Proposition 1.3.2 (Young integral). Let $X \in \mathcal{C}^\alpha([0,T];V)$ and $Y \in \mathcal{C}^\beta([0,T];\mathcal{L}(V,W))$, where

$\alpha, \beta \in (0,1]$ and $\alpha + \beta > 1$. Then

$$\int_0^t Y_u \, \mathrm{d}X_u := \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_u X_{u,v}$$

exists and satisfies

$$\left| \int_s^t Y_u \, \mathrm{d}X_u - Y_s X_{s,t} \right| \le C_{\alpha+\beta} \|Y\|_\beta \|X\|_\alpha |t-s|^{\alpha+\beta}.$$

Proof. Set $A_{s,t} = Y_s X_{s,t}$. Then $\delta A_{s,u,t} = -Y_{s,u} X_{u,t}$, so

$$|\delta A_{s,u,t}| \le \|Y\|_\beta \|X\|_\alpha |t-s|^{\alpha+\beta}.$$

Apply the sewing lemma. □

Proposition 1.3.3 (Stability of Young integrals). Under the preceding assumptions, let also $\widetilde{X} \in \mathcal{C}^\alpha([0,T]; V)$ and $\widetilde{Y} \in \mathcal{C}^\beta([0,T]; \mathcal{L}(V,W))$. Then

$$\begin{aligned} &\left\| \int_0^\cdot Y_u \, \mathrm{d}X_u - \int_0^\cdot \widetilde{Y}_u \, \mathrm{d}\widetilde{X}_u \right\|_\alpha \\ &\quad \le C \Big[\big(|Y_0 - \widetilde{Y}_0| + \|Y - \widetilde{Y}\|_\beta \big) \|X\|_\alpha + \big(|\widetilde{Y}_0| + \|\widetilde{Y}\|_\beta \big) \|X - \widetilde{X}\|_\alpha \Big], \end{aligned}$$

where C depends only on α, β, and T.

Proof. Use bilinearity to write the difference as

$$\int_0^\cdot (Y - \widetilde{Y})_u \, \mathrm{d}X_u + \int_0^\cdot \widetilde{Y}_u \, \mathrm{d}(X - \widetilde{X})_u.$$

For $s < t$, the Young estimate gives

$$\begin{aligned} \left| \int_s^t (Y - \widetilde{Y})_u \, \mathrm{d}X_u \right| &\le |Y_s - \widetilde{Y}_s| \, |X_{s,t}| + C \|Y - \widetilde{Y}\|_\beta \|X\|_\alpha |t-s|^{\alpha+\beta}, \\ \left| \int_s^t \widetilde{Y}_u \, \mathrm{d}(X - \widetilde{X})_u \right| &\le |\widetilde{Y}_s| \, |(X - \widetilde{X})_{s,t}| \\ &\quad + C \|\widetilde{Y}\|_\beta \|X - \widetilde{X}\|_\alpha |t-s|^{\alpha+\beta}. \end{aligned}$$

Bound the two supremum norms by their initial values plus T^β times the corresponding Hölder seminorms, divide by $|t-s|^\alpha$, and take the supremum. □

Lemma 1.3.4 (Young integration by parts). Let $B \colon V \times W \to U$ be a continuous bilinear map, $X \in \mathcal{C}^\alpha([0,T]; V)$, and $Y \in \mathcal{C}^\beta([0,T]; W)$, with $\alpha + \beta > 1$. Then

$$B(X_T, Y_T) - B(X_0, Y_0) = \int_0^T B(X_u, \mathrm{d}Y_u) + \int_0^T B(\mathrm{d}X_u, Y_u).$$

For scalar-valued paths this is the usual formula $X_T Y_T - X_0 Y_0 = \int X \, \mathrm{d}Y + \int Y \, \mathrm{d}X$.

Proof. For every partition,

$$B(X_t, Y_t) - B(X_s, Y_s) = B(X_s, Y_{s,t}) + B(X_{s,t}, Y_s) + B(X_{s,t}, Y_{s,t}).$$

The sum of the last terms is bounded by $C\|X\|_\alpha \|Y\|_\beta T|\pi|^{\alpha+\beta-1}$ and tends to zero. The first two sums converge to the two Young integrals. $\square$

For Banach spaces V, W, let $C^{k+\gamma}(V; W)$ denote the maps that are k times continuously Fréchet differentiable and whose kth derivative is locally γ-Hölder.

Lemma 1.3.5 (Young chain rule). Let $X \in \mathcal{C}^\alpha([0,T]; V)$ and let $\gamma \in (0,1]$. If $f \in C^{1+\gamma}(V; W)$ and $\alpha(1+\gamma) > 1$, then $u \mapsto Df(X_u)$ is $\alpha\gamma$-Hölder on $[0,T]$, the Young integral is well defined, and

$$f(X_T) - f(X_0) = \int_0^T Df(X_u)\,\mathrm{d}X_u.$$

Proof. The compactness of $X([0,T])$ supplies a local Hölder constant for Df, so $\|Df(X)\|_{\alpha\gamma} < \infty$. Taylor's formula gives

$$f(X_t) - f(X_s) = Df(X_s)X_{s,t} + R_{s,t}, \qquad |R_{s,t}| \lesssim |X_{s,t}|^{1+\gamma} \lesssim |t-s|^{\alpha(1+\gamma)}.$$

Sum over a partition. Since $\alpha(1+\gamma) > 1$, the remainder sum tends to zero, while the principal sum converges to the Young integral. $\square$

Lemma 1.3.6 (Associativity of Young integration). Let

$$X \in \mathcal{C}^\alpha([0,T]; V), \quad K \in \mathcal{C}^\beta([0,T]; \mathcal{L}(V,U)), \quad Y \in \mathcal{C}^\beta([0,T]; \mathcal{L}(U,W)),$$

with $\alpha + \beta > 1$, and put $Z_t := \int_0^t K_u\,\mathrm{d}X_u$. Then $Z \in \mathcal{C}^\alpha([0,T]; U)$ and

$$\int_0^T Y_u\,\mathrm{d}Z_u = \int_0^T (Y_u \circ K_u)\,\mathrm{d}X_u.$$

Proof. The Young estimate shows that $Z_{s,t} = K_s X_{s,t} + O(|t-s|^{\alpha+\beta})$, uniformly in s,t, and hence $Z \in \mathcal{C}^\alpha$. Consequently,

$$\int_s^t Y_u\,\mathrm{d}Z_u = Y_s K_s X_{s,t} + O(|t-s|^{\alpha+\beta}) = \int_s^t (Y_u \circ K_u)\,\mathrm{d}X_u + O(|t-s|^{\alpha+\beta}).$$

Summing over a partition makes the total error tend to zero because $\alpha + \beta > 1$. $\square$

1.4 Rough Integration

Definition 1.4.1 (Controlled path). Let $\alpha \in (1/3, 1/2]$ and $X \in \mathcal{C}^\alpha([0,T];V)$. A controlled path over X with values in W is a pair

$$Y \in \mathcal{C}^\alpha([0,T];W), \qquad Y' \in \mathcal{C}^\alpha([0,T];\mathcal{L}(V,W)),$$

for which the remainder

$$R^Y_{s,t} := Y_{s,t} - Y'_s X_{s,t}$$

belongs to $\mathcal{C}^{2\alpha}_2(\Delta_{[0,T]};W)$. The space of such pairs is denoted by $\mathscr{D}^{2\alpha}_X([0,T];W)$, and Y' is called the Gubinelli derivative.

Define

$$\|(Y,Y')\|_{X,2\alpha} := \|Y'\|_\alpha + \|R^Y\|_{2\alpha}.$$

Then

$$|Y_0| + |Y'_0| + \|(Y,Y')\|_{X,2\alpha}$$

is a Banach norm on $\mathscr{D}^{2\alpha}_X$. Moreover,

$$\|Y\|_\alpha \le \|Y'\|_\infty \|X\|_\alpha + T^\alpha \|R^Y\|_{2\alpha} \lesssim_{T,X} |Y'_0| + \|(Y,Y')\|_{X,2\alpha}.$$

Remark. The Gubinelli derivative need not be unique for a very regular driver. For example, if both X and Y are 2α-Hölder, then every $Y' \in \mathcal{C}^\alpha([0,T];\mathcal{L}(V,W))$ produces a 2α-Hölder remainder. Merely assuming that Y' is continuous is insufficient, because the definition also requires Y' to be α-Hölder.

Example (Products of controlled paths). Let $B\colon W_1 \times W_2 \to W_3$ be continuous bilinear, and let $(Y,Y') \in \mathscr{D}^{2\alpha}_X(W_1)$ and $(Z,Z') \in \mathscr{D}^{2\alpha}_X(W_2)$. Then $B(Y,Z)$ is controlled by X with derivative

$$B(Y,Z)'_s(v) = B(Y_s, Z'_s v) + B(Y'_s v, Z_s).$$

Indeed,

$$R^{B(Y,Z)}_{s,t} = B(Y_s, R^Z_{s,t}) + B(R^Y_{s,t}, Z_s) + B(Y_{s,t}, Z_{s,t}),$$

which is 2α-Hölder.

Proposition 1.4.2 (Rough integral). Let

$$\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T];V)$$

and suppose

$$(Y,Y') \in \mathscr{D}^{2\alpha}_X([0,T];\mathcal{L}(V,W)).$$

Here Y'_s is viewed as a continuous bilinear map on $V \times V$ when contracted with $\mathbb{X}_{s,t}$. Then

$$\int_0^t Y_u \, \mathrm{d}\mathbf{X}_u := \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} \left(Y_u X_{u,v} + Y'_u \mathbb{X}_{u,v} \right)$$

exists, and

$$\left| \int_s^t Y_u \, \mathrm{d}\mathbf{X}_u - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \le C_\alpha \left(\|R^Y\|_{2\alpha} \|X\|_\alpha + \|Y'\|_\alpha \|\mathbb{X}\|_{2\alpha} \right) |t-s|^{3\alpha}.$$

Proof. Set $A_{s,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}$. Chen's relation and the controlled expansion give

$$\begin{aligned} \delta A_{s,u,t} &= -Y_{s,u} X_{u,t} + Y'_s (X_{s,u} \otimes X_{u,t}) - Y'_{s,u} \mathbb{X}_{u,t} \\ &= -R^Y_{s,u} X_{u,t} - Y'_{s,u} \mathbb{X}_{u,t}. \end{aligned}$$

Therefore

$$|\delta A_{s,u,t}| \le \left(\|R^Y\|_{2\alpha} \|X\|_\alpha + \|Y'\|_\alpha \|\mathbb{X}\|_{2\alpha} \right) |t-s|^{3\alpha}.$$

Since $3\alpha > 1$, the sewing lemma applies. □

Proposition 1.4.3 (The rough integral is controlled). With the preceding assumptions, put

$$Z_t := \int_0^t Y_u \, \mathrm{d}\mathbf{X}_u, \qquad Z'_t := Y_t.$$

Then $(Z, Z') \in \mathscr{D}_X^{2\alpha}([0,T]; W)$ and

$$\begin{aligned} \|R^Z\|_{2\alpha} &\le \|Y'\|_\infty \|\mathbb{X}\|_{2\alpha} \\ &\quad + C_\alpha T^\alpha \left(\|R^Y\|_{2\alpha} \|X\|_\alpha + \|Y'\|_\alpha \|\mathbb{X}\|_{2\alpha} \right). \end{aligned}$$

Proof. The rough-integral estimate yields

$$R^Z_{s,t} = Z_{s,t} - Y_s X_{s,t} = Y'_s \mathbb{X}_{s,t} + O(|t-s|^{3\alpha}),$$

which is 2α-Hölder. □

For two controlled paths over possibly different drivers, define the mixed seminorm

$$\|(Y, Y'); (\tilde{Y}, \tilde{Y}')\|_{X, \tilde{X}; 2\alpha} := \|Y' - \tilde{Y}'\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha}.$$

Theorem 1.4.4 (Stability of controlled paths and rough-integral remainders). Let $\mathbf{X} = (X, \mathbb{X})$ and $\tilde{\mathbf{X}} = (\tilde{X}, \tilde{\mathbb{X}})$ be α-Hölder rough paths. Let $(Y, Y') \in \mathscr{D}_X^{2\alpha}$ and $(\tilde{Y}, \tilde{Y}') \in \mathscr{D}_{\tilde{X}}^{2\alpha}$. Then

$$\begin{aligned} \|Y - \tilde{Y}\|_\alpha \le C_T \Big[&(|Y'_0 - \tilde{Y}'_0| + \|Y' - \tilde{Y}'\|_\alpha) \|X\|_\alpha \\ &+ (|\tilde{Y}'_0| + \|\tilde{Y}'\|_\alpha) \|X - \tilde{X}\|_\alpha + T^\alpha \|R^Y - R^{\tilde{Y}}\|_{2\alpha} \Big]. \end{aligned}$$

If $Z = \int Y \, \mathrm{d}\mathbf{X}$ and $\tilde{Z} = \int \tilde{Y} \, \mathrm{d}\tilde{\mathbf{X}}$, then

$$\|R^Z - R^{\tilde{Z}}\|_{2\alpha} \le C_{\alpha,T} \Big[(|Y'_0 - \tilde{Y}'_0| + \|Y' - \tilde{Y}'\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha}) \|\mathbf{X}\|_\alpha$$

$$+\,(|\widetilde{Y}_0'| + \|\widetilde{Y}'\|_\alpha + \|R^{\widetilde{Y}}\|_{2\alpha})\|\mathbf{X};\widetilde{\mathbf{X}}\|_\alpha\Big].$$

Proof. The first estimate follows from

$$(Y-\widetilde{Y})_{s,t} = (Y_s' - \widetilde{Y}_s')X_{s,t} + \widetilde{Y}_s'(X-\widetilde{X})_{s,t} + (R^Y - R^{\widetilde{Y}})_{s,t}.$$

For the second, set

$$A_{s,t} = Y_s X_{s,t} + Y_s'\mathbb{X}_{s,t}, \qquad \widetilde{A}_{s,t} = \widetilde{Y}_s\widetilde{X}_{s,t} + \widetilde{Y}_s'\widetilde{\mathbb{X}}_{s,t}.$$

The difference of the sewing defects is

$$\begin{aligned}\delta(A-\widetilde{A})_{s,u,t} = &-(R^Y - R^{\widetilde{Y}})_{s,u}X_{u,t} - R^{\widetilde{Y}}_{s,u}(X-\widetilde{X})_{u,t} \\ &-(Y'-\widetilde{Y}')_{s,u}\mathbb{X}_{u,t} - \widetilde{Y}'_{s,u}(\mathbb{X}-\widetilde{\mathbb{X}})_{u,t}.\end{aligned}$$

Each term is bounded by the displayed coefficients times $|t-s|^{3\alpha}$. The sewing lemma gives a corresponding estimate for the difference between the two rough integrals and their local models. Finally,

$$Y_s'\mathbb{X}_{s,t} - \widetilde{Y}_s'\widetilde{\mathbb{X}}_{s,t} = (Y_s' - \widetilde{Y}_s')\mathbb{X}_{s,t} + \widetilde{Y}_s'(\mathbb{X}-\widetilde{\mathbb{X}})_{s,t},$$

which yields the stated 2α-Hölder remainder bound. □

Corollary 1.4.5 (Stability of the rough integral). Under the assumptions of the theorem,

$$\begin{aligned}\left\|\int_0^\cdot Y_u\,\mathrm{d}\mathbf{X}_u - \int_0^\cdot \widetilde{Y}_u\,\mathrm{d}\widetilde{\mathbf{X}}_u\right\|_\alpha \le C\Big[&\|Y-\widetilde{Y}\|_\infty\|X\|_\alpha + \|\widetilde{Y}\|_\infty\|X-\widetilde{X}\|_\alpha \\ &+ T^\alpha\|R^Z - R^{\widetilde{Z}}\|_{2\alpha}\Big],\end{aligned}$$

where $Z = \int Y\,\mathrm{d}\mathbf{X}$ and $\widetilde{Z} = \int \widetilde{Y}\,\mathrm{d}\widetilde{\mathbf{X}}$. Combining this with the preceding theorem and the controlled-path bounds yields local Lipschitz continuity of rough integration in the driver and integrand.

Proof. Since (Z,Y) is controlled by X and $(\widetilde{Z},\widetilde{Y})$ is controlled by $\widetilde{X}$,

$$(Z-\widetilde{Z})_{s,t} = (Y_s - \widetilde{Y}_s)X_{s,t} + \widetilde{Y}_s(X-\widetilde{X})_{s,t} + (R^Z - R^{\widetilde{Z}})_{s,t}.$$

Divide by $|t-s|^\alpha$, take the supremum, and use the estimates above. □

Chapter 2 Properties of Rough Integration

2.1 Associativity

We first explain how to integrate with respect to a controlled path. All products and compositions below are assumed to have compatible Banach-space types.

Definition 2.1.1 (Integration with respect to a rough integral). Let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha$ and let $(Y, Y'), (K, K') \in \mathscr{D}_X^{2\alpha}$. Put

$$Z_t := \int_0^t K_u \, \mathrm{d}\mathbf{X}_u, \qquad Z'_t := K_t.$$

Then $(Z, Z') \in \mathscr{D}_X^{2\alpha}$. We define

$$\int_0^\cdot Y_u \, \mathrm{d}Z_u := \int_0^\cdot (Y_u K_u) \, \mathrm{d}\mathbf{X}_u,$$

where the right-hand side is the rough integral of the controlled product YK.

Proposition 2.1.2 (Riemann-sum representation). Under the preceding assumptions, for every $t \in [0, T]$,

$$\int_0^t Y_u \, \mathrm{d}Z_u = \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} \left(Y_u Z_{u,v} + Y'_u Z'_u \mathbb{X}_{u,v} \right).$$

Proof. The rough-integral expansion gives

$$Z_{s,t} = K_s X_{s,t} + K'_s \mathbb{X}_{s,t} + O(|t-s|^{3\alpha}),$$

and

$$\int_s^t (Y_u K_u) \, \mathrm{d}\mathbf{X}_u = Y_s K_s X_{s,t} + (YK)'_s \mathbb{X}_{s,t} + O(|t-s|^{3\alpha}).$$

Since $(YK)' = YK' + Y'K$,

$$\begin{aligned} Y_s Z_{s,t} + Y'_s Z'_s \mathbb{X}_{s,t} &= Y_s K_s X_{s,t} + (Y_s K'_s + Y'_s K_s) \mathbb{X}_{s,t} \\ &= \int_s^t (Y_u K_u) \, \mathrm{d}\mathbf{X}_u + O(|t-s|^{3\alpha}). \end{aligned}$$

The sum of the errors over a partition tends to zero because $3\alpha > 1$. $\square$

More generally, the displayed Riemann sums can be used to define integration of one controlled path against another, even when the integrator is not itself a rough integral.

Let $(Z, Z') \in \mathscr{D}_X^{2\alpha}([0,T]; W)$. Define its canonical second level by

$$\mathbb{Z}_{s,t} := \int_s^t Z_{s,u} \otimes \mathrm{d}Z_u.$$

This is an integral of controlled paths. It satisfies Chen's relation and is 2α-Hölder, so $\mathbf{Z} = (Z, \mathbb{Z})$ is a rough path. Locally,

$$\mathbb{Z}_{s,t} = (Z'_s \otimes Z'_s)\mathbb{X}_{s,t} + O(|t-s|^{3\alpha}).$$

Lemma 2.1.3 (Compatibility with the canonical lift). Let $\mathbf{Z} = (Z, \mathbb{Z})$ be the canonical lift just described, and let $(Y, Y') \in \mathscr{D}_Z^{2\alpha}$. Then $(Y, Y'Z') \in \mathscr{D}_X^{2\alpha}$ and

$$\int_0^\cdot Y_u \,\mathrm{d}\mathbf{Z}_u = \int_0^\cdot Y_u \,\mathrm{d}Z_u,$$

where the right-hand side is the controlled-path integral relative to X.

Proof. Since

$$Y_{s,t} = Y'_s Z_{s,t} + R^Y_{s,t} = Y'_s Z'_s X_{s,t} + Y'_s R^Z_{s,t} + R^Y_{s,t},$$

Y is controlled by X with derivative $Y'Z'$. Moreover,

$$\begin{aligned}\int_s^t Y_u \,\mathrm{d}\mathbf{Z}_u &= Y_s Z_{s,t} + Y'_s \mathbb{Z}_{s,t} + O(|t-s|^{3\alpha}) \\ &= Y_s Z_{s,t} + Y'_s (Z'_s \otimes Z'_s)\mathbb{X}_{s,t} + O(|t-s|^{3\alpha}),\end{aligned}$$

which is the local model for the controlled-path integral of Y against Z. Summing over partitions proves equality. □

2.2 The Rough Itô Formula

Definition 2.2.1 (Bracket). For $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha$, define

$$[\mathbf{X}]_t := X_{0,t} \otimes X_{0,t} - 2\,\mathrm{Sym}(\mathbb{X}_{0,t}).$$

Then

$$[\mathbf{X}]_{s,t} = X_{s,t} \otimes X_{s,t} - 2\,\mathrm{Sym}(\mathbb{X}_{s,t}),$$

so $[\mathbf{X}] \in \mathcal{C}^{2\alpha}([0,T]; V \otimes V)$.

A rough path is weakly geometric exactly when $[\mathbf{X}] \equiv 0$.

Lemma 2.2.2 (Bracket of a rough integral). Let $(K, K') \in \mathscr{D}_X^{2\alpha}$ and set $Z = \int K \,\mathrm{d}\mathbf{X}$. If $\mathbf{Z}$ is the canonical lift of Z, then

$$[\mathbf{Z}]_t = \int_0^t (K_u \otimes K_u)\,\mathrm{d}[\mathbf{X}]_u,$$

where the right-hand side is a Young integral.

Proof. The Young integral exists because $K \otimes K$ is α-Hölder and $[\mathbf{X}]$ is 2α-Hölder. The local expansions for Z and $\mathbb{Z}$ imply

$$[\mathbf{Z}]_{s,t} = (K_s \otimes K_s)[\mathbf{X}]_{s,t} + O(|t-s|^{3\alpha}).$$

Summing over a partition proves the identity. □

Theorem 2.2.3 (Rough Itô formula). Let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T]; V)$ and let $f\colon V \to W$ be C^3 on an open neighborhood of all line segments joining points of $X([0,T])$, with bounded derivatives there. Then

$$f(X_T) = f(X_0) + \int_0^T Df(X_u)\,\mathrm{d}\mathbf{X}_u + \frac{1}{2}\int_0^T D^2 f(X_u)\,\mathrm{d}[\mathbf{X}]_u.$$

The first integral is rough and the second is Young.

Proof. Taylor's theorem gives, uniformly in $s < t$,

$$f(X_t) - f(X_s) = Df(X_s)X_{s,t} + \frac{1}{2}D^2 f(X_s)[X_{s,t}, X_{s,t}] + O(|t-s|^{3\alpha}).$$

Because $D^2 f(X_s)$ is symmetric, $D^2 f(X_s)\mathbb{X}_{s,t}$ depends only on $\mathrm{Sym}(\mathbb{X}_{s,t})$. Hence

$$\frac{1}{2}D^2 f(X_s)[X_{s,t}, X_{s,t}] = D^2 f(X_s)\mathbb{X}_{s,t} + \frac{1}{2}D^2 f(X_s)[\mathbf{X}]_{s,t}.$$

The first two local terms are the rough-integral model for $Df(X)$, while the last is the Young-integral model against the bracket. Sum over partitions; the $O(|t-s|^{3\alpha})$ terms vanish because $3\alpha > 1$. □

Corollary 2.2.4 (Itô formula for a controlled dynamics). Suppose $(Y, Y') \in \mathscr{D}_X^{2\alpha}$ and $(Y', Y'') \in \mathscr{D}_X^{2\alpha}$, and suppose

$$Y_t = Y_0 + \int_0^t Y'_u\,\mathrm{d}\mathbf{X}_u + \Gamma_t, \qquad \Gamma \in \mathcal{C}^{2\alpha}([0,T]).$$

If f is C^3 with the required derivative bounds along the range of Y, then

$$\begin{aligned} f(Y_T) = f(Y_0) &+ \int_0^T Df(Y_u)Y'_u\,\mathrm{d}\mathbf{X}_u + \int_0^T Df(Y_u)\,\mathrm{d}\Gamma_u \\ &+ \frac{1}{2}\int_0^T D^2 f(Y_u)[Y'_u \otimes Y'_u]\,\mathrm{d}[\mathbf{X}]_u. \end{aligned}$$

Proof. The rough-integral expansion gives

$$Y_{s,t} = Y'_s X_{s,t} + Y''_s \mathbb{X}_{s,t} + \Gamma_{s,t} + O(|t-s|^{3\alpha}).$$

Taylor-expand $f(Y_t) - f(Y_s)$ to second order. The local model for the first rough integral is

$$Df(Y_s)Y'_s X_{s,t} + \big(D^2 f(Y_s)[Y'_s, Y'_s] + Df(Y_s)Y''_s\big)\mathbb{X}_{s,t}.$$

The Young integral against Γ contributes $Df(Y_s)\Gamma_{s,t}$, and the bracket term contributes

$$\frac{1}{2}D^2 f(Y_s)[Y'_s, Y'_s]\big(X_{s,t}^{\otimes 2} - 2\,\mathrm{Sym}\,\mathbb{X}_{s,t}\big).$$

Since $D^2 f$ is symmetric, the bracket term cancels precisely the symmetric second-level part and leaves the quadratic Taylor term. The remaining error is $O(|t-s|^{3\alpha})$; summing over partitions proves the formula. □

2.3 Composition of Controlled Paths

Let $\varphi\colon W \to \overline{W}$ be a smooth map and $(Y, Y') \in \mathscr{D}_X^{2\alpha}([0,T]; W)$. Define

$$\varphi(Y)'_t := D\varphi(Y_t)Y'_t.$$

Lemma 2.3.1 (Estimate for a composition). Assume $\varphi \in C_b^2(W; \overline{W})$ and $|Y'_0| + \|(Y, Y')\|_{X,2\alpha} \leq M$ with $M \geq 1$. Then $(\varphi(Y), D\varphi(Y)Y') \in \mathscr{D}_X^{2\alpha}$ and

$$\|(\varphi(Y), D\varphi(Y)Y')\|_{X,2\alpha} \leq C_{\alpha,T} M \|\varphi\|_{C_b^2}(1 + \|X\|_\alpha)^2\big(|Y'_0| + \|(Y, Y')\|_{X,2\alpha}\big).$$

For $T \leq 1$, the constant can be chosen independently of T.

Proof. The product rule gives

$$\|D\varphi(Y)Y'\|_\alpha \leq \|D\varphi(Y)\|_\infty \|Y'\|_\alpha + \|Y'\|_\infty \|D^2\varphi(Y)\|_\infty \|Y\|_\alpha.$$

Taylor's formula yields

$$R_{s,t}^{\varphi(Y)} = \varphi(Y_t) - \varphi(Y_s) - D\varphi(Y_s)Y_{s,t} + D\varphi(Y_s)R_{s,t}^Y,$$

so

$$\|R^{\varphi(Y)}\|_{2\alpha} \leq \tfrac{1}{2}\|D^2\varphi\|_\infty \|Y\|_\alpha^2 + \|D\varphi\|_\infty \|R^Y\|_{2\alpha}.$$

Insert the controlled-path bound for $\|Y\|_\alpha$ and collect terms. □

Remark. Global boundedness of the derivatives is a convenient sufficient assumption, not a consequence of boundedness of the path. In applications, it is enough to bound the derivatives on the compact set of line segments joining points in $Y([0,T])$ (and the analogous compact sets used by higher-order Taylor remainders). No global extension of φ is being assumed.

Lemma 2.3.2 (Stability of composition on Hölder paths). Let $\varphi \in C_b^2(W; \overline{W})$, $T \leq 1$, and $X, Y \in \mathcal{C}^\alpha([0,T]; W)$ with $\|X\|_\alpha, \|Y\|_\alpha \leq K$. Then

$$\|\varphi(X) - \varphi(Y)\|_\alpha \leq C_{\alpha,K}\|\varphi\|_{C_b^2}\big(|X_0 - Y_0| + \|X - Y\|_\alpha\big).$$

Proof. Write

$$\varphi(x) - \varphi(y) = g(x,y)(x-y), \qquad g(x,y) := \int_0^1 D\varphi(y + \theta(x-y))\, d\theta.$$

Then $\|g\|_\infty \le \|D\varphi\|_\infty$ and g is Lipschitz with constant controlled by $\|D^2\varphi\|_\infty$. Apply this identity at times s and t, subtract, and use $\|X - Y\|_\infty \le |X_0 - Y_0| + \|X - Y\|_\alpha$ because $T \le 1$. $\square$

For paths controlled by different drivers, set

$$\|(Y,Y');(\widetilde{Y},\widetilde{Y}')\|_{X,\widetilde{X};2\alpha} := \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha}.$$

Theorem 2.3.3 (Stability of composition on controlled paths). Let $T \le 1$, let $(Y,Y') \in \mathscr{D}_X^{2\alpha}$ and $(\widetilde{Y},\widetilde{Y}') \in \mathscr{D}_{\widetilde{X}}^{2\alpha}$, and let $\varphi \in C_b^3$. Assume

$$\|X\|_\alpha + \|\widetilde{X}\|_\alpha + |Y_0'| + |\widetilde{Y}_0'| + \|(Y,Y')\|_{X,2\alpha} + \|(\widetilde{Y},\widetilde{Y}')\|_{\widetilde{X},2\alpha} \le M.$$

Define $Z = \varphi(Y)$, $Z' = D\varphi(Y)Y'$ and similarly $\widetilde{Z}, \widetilde{Z}'$. Then

$$\begin{aligned}
&\|(Z,Z');(\widetilde{Z},\widetilde{Z}')\|_{X,\widetilde{X};2\alpha} + \|Z - \widetilde{Z}\|_\alpha \\
&\quad \le C_M\Big(\|X - \widetilde{X}\|_\alpha + |Y_0 - \widetilde{Y}_0| + |Y_0' - \widetilde{Y}_0'| + \|(Y,Y');(\widetilde{Y},\widetilde{Y}')\|_{X,\widetilde{X};2\alpha}\Big),
\end{aligned}$$

where C_M depends on M, α, and the C_b^3 bounds of φ.

Proof. For the derivative, add and subtract $D\varphi(Y)\widetilde{Y}'$ and use the preceding Hölder composition lemma:

$$D\varphi(Y)Y' - D\varphi(\widetilde{Y})\widetilde{Y}' = D\varphi(Y)(Y' - \widetilde{Y}') + (D\varphi(Y) - D\varphi(\widetilde{Y}))\widetilde{Y}'.$$

For the remainder, use the exact Taylor representation

$$\begin{aligned}
R_{s,t}^{\varphi(Y)} &= \int_0^1 (1-\theta) D^2\varphi(Y_s + \theta Y_{s,t})[Y_{s,t}, Y_{s,t}]\, d\theta \\
&\quad + D\varphi(Y_s) R_{s,t}^Y,
\end{aligned}$$

and the analogous identity for $\widetilde{Y}$. Add and subtract the same terms with Y replaced by $\widetilde{Y}$ one factor at a time. The C_b^3 bound controls the change of $D^2\varphi$, while the controlled-path stability estimate controls $\|Y - \widetilde{Y}\|_\alpha$. Dividing by $|t-s|^{2\alpha}$ gives the first bound; the second follows from the controlled decompositions. $\square$

2.4 Fubini's Theorem and Indefinite Rough Integrals

Theorem 2.4.1 (Rough Fubini theorem). Let $\mathbf{X} \in \mathscr{C}^\alpha([0,T];V)$ with $\alpha \in (1/3, 1/2]$, and let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Suppose

$$\omega \longmapsto (Y^\omega, (Y^\omega)')$$

is Bochner integrable as a map into the Banach space $\mathscr{D}_X^{2\alpha}([0,T];\mathcal{L}(V,W))$; equivalently, it is strongly measurable and

$$\int_\Omega \Big(|Y_0^\omega| + |(Y_0^\omega)'| + \|(Y^\omega,(Y^\omega)')\|_{X,2\alpha} \Big)\, \mu(\mathrm{d}\omega) < \infty.$$

Then the pointwise Bochner integral is a controlled path and

$$\int_\Omega \left(\int_0^T Y_u^\omega \,\mathrm{d}\mathbf{X}_u \right) \mu(\mathrm{d}\omega) = \int_0^T \left(\int_\Omega Y_u^\omega \,\mu(\mathrm{d}\omega) \right) \mathrm{d}\mathbf{X}_u.$$

The Gubinelli derivative of the integrated path is $\int_\Omega (Y^\omega)'\,\mathrm{d}\mu$, and its remainder is $\int_\Omega R^{Y^\omega}\,\mathrm{d}\mu$.

Proof. The controlled-path space is Banach, and rough integration is a bounded linear map from that space into $\mathcal{C}^\alpha([0,T];W)$. Every bounded linear map commutes with Bochner integration. The derivative and remainder formulas follow by integrating $Y_{s,t}^\omega = (Y_s^\omega)' X_{s,t} + R_{s,t}^{Y^\omega}$. $\square$

Proposition 2.4.2 (Singular Hölder paths). Let $0 < \alpha \le 1$, $\eta \le \alpha$, and let Y be defined on $(0,T]$. Then

$$\|Y\|_{\alpha,\eta} := \sup_{0<s<t\le T} \frac{|Y_t - Y_s|}{s^{\eta-\alpha}|t-s|^\alpha} < \infty$$

if and only if

$$\|Y\|_{\alpha;[\varepsilon,T]} = O(\varepsilon^{\eta-\alpha}) \qquad (\varepsilon \downarrow 0).$$

The resulting class is denoted by $\mathcal{C}^{\alpha,\eta}((0,T])$.

Proof. If the weighted seminorm is finite, then for $s \ge \varepsilon$, $s^{\eta-\alpha} \le \varepsilon^{\eta-\alpha}$ because $\eta - \alpha \le 0$, which gives the local estimate. Conversely, if $\|Y\|_{\alpha;[\varepsilon,T]} \le C\varepsilon^{\eta-\alpha}$, choose $\varepsilon = s$ to obtain $|Y_t - Y_s| \le Cs^{\eta-\alpha}|t-s|^\alpha$ directly. $\square$

Theorem 2.4.3 (Improper Young integral). Let $X \in \mathcal{C}^\alpha([0,T])$ with $\alpha > 1/2$, and let $Y \in \mathcal{C}^{\alpha,\eta}((0,T])$, where $\eta \le \alpha$, $\eta + \alpha > 0$, and $\eta \ne 0$. Then

$$Z_t := \int_{0+}^t Y\,\mathrm{d}X := \lim_{\varepsilon\downarrow 0} \int_\varepsilon^t Y\,\mathrm{d}X$$

exists and belongs to $\mathcal{C}^{\alpha,\alpha+\eta\wedge 0}((0,T])$.

Proof. A dyadic telescoping argument using $|Y_{2s} - Y_s| \lesssim s^\eta$ gives

$$|Y_s| \lesssim 1 + s^{\eta\wedge 0}, \qquad 0 < s \le T.$$

Young's estimate on a dyadic interval $[s,2s]$ gives

$$\begin{aligned} \left| \int_s^{2s} Y\,\mathrm{d}X \right| &\lesssim |Y_s| s^\alpha + \|Y\|_{\alpha;[s,T]} s^{2\alpha} \\ &\lesssim s^{\alpha+\eta\wedge 0} + s^{\alpha+\eta}. \end{aligned}$$

Both exponents are positive under $\eta + \alpha > 0$, so the dyadic tail is summable and the improper integral exists.

To obtain the weighted Hölder estimate, first suppose $h = t - s \le s$. Then

$$|Z_{s,t}| \lesssim s^{\eta\wedge 0} h^{\alpha} + s^{\eta-\alpha} h^{2\alpha} \lesssim s^{\eta\wedge 0} h^{\alpha}.$$

If $h > s$, decompose $[s,t]$ into dyadic intervals until their size is comparable to h. Summing the preceding dyadic bounds gives $|Z_{s,t}| \lesssim h^{\alpha+\eta\wedge 0}$, which is at most $Cs^{\eta\wedge 0}h^{\alpha}$ when $\eta < 0$ and is Ch^{α} when $\eta > 0$. This is exactly the $\mathcal{C}^{\alpha,\alpha+\eta\wedge 0}$ estimate. □

Proposition 2.4.4 (Singular controlled paths). Let $\mathbf{X} \in \mathscr{C}^{\alpha}([0,T])$ with $\alpha \in (1/3, 1/2]$, and let (Y, Y') be defined on $(0,T]$. For $\eta \le 2\alpha$, the condition

$$\|(Y,Y')\|_{X,2\alpha;[\varepsilon,T]} = O(\varepsilon^{\eta-2\alpha})$$

is equivalent to finiteness of

$$\|(Y,Y')\|_{X,2\alpha,\eta} := \sup_{0<s<t\le T} \frac{|Y'_t - Y'_s|}{s^{\eta-2\alpha}|t-s|^{\alpha}} + \sup_{0<s<t\le T} \frac{|Y_t - Y_s - Y'_s X_{s,t}|}{s^{\eta-2\alpha}|t-s|^{2\alpha}}.$$

The corresponding class is denoted by $\mathscr{D}_X^{2\alpha,\eta}$.

Proof. The proof is the same direct comparison as for singular Hölder paths. In one direction, use $s \ge \varepsilon$ and $\eta - 2\alpha \le 0$. In the other, choose $\varepsilon = s$ in the assumed local estimate. The suprema must start at $s > 0$ because the weights may be singular at the origin. □

Theorem 2.4.5 (Improper rough integral). Let $\mathbf{X} \in \mathscr{C}^{\alpha}([0,T])$ with $\alpha \in (1/3, 1/2]$, and let $(Y, Y') \in \mathscr{D}_X^{2\alpha,\eta}$, where $-\alpha < \eta \le 2\alpha$ and $\eta \ne 0$. Then

$$Z_t := \int_{0^+}^{t} Y \, d\mathbf{X} := \lim_{\varepsilon\downarrow 0} \int_{\varepsilon}^{t} Y \, d\mathbf{X}$$

exists,

$$Z \in \mathcal{C}^{\alpha,\alpha+\eta\wedge 0}((0,T]),$$

and, more precisely,

$$(Z, Z') := \left(\int_{0^+}^{\cdot} Y \, d\mathbf{X}, \, Y \right) \in \mathscr{D}_X^{2\alpha,\alpha+\eta\wedge 0}.$$

Proof. The weighted controlled estimates imply, by dyadic telescoping,

$$|Y_s| \lesssim 1 + s^{\eta\wedge 0},$$

and

$$|Y'_s| \lesssim \begin{cases} 1 + s^{\eta-\alpha}, & \eta < \alpha, \\ 1 + |\log s|, & \eta = \alpha, \\ 1, & \eta > \alpha. \end{cases}$$

In particular, the coarser bound $|Y'_s| \lesssim s^{\eta\wedge 0-\alpha}$ holds on $(0, T]$. The rough-integral estimate on $[s, 2s]$ therefore gives a contribution of order $s^{\alpha+\eta\wedge 0}$ from $Y_s X_{s,2s}$, a contribution of order $s^{2\alpha+\eta\wedge 0-\alpha}$ from $Y'_s \mathbb{X}_{s,2s}$, and a remainder of order $s^{\alpha+\eta}$. At the borderline $\eta = \alpha$, the second contribution is instead $O(s^{2\alpha}(1 + |\log s|))$. All these dyadic terms are summable when $\eta > -\alpha$, so the improper integral exists.

Let $h = t - s \leq s$. The controlled relation and the weighted bounds give

$$|Y_{s,t}| \lesssim s^{\eta\wedge 0-\alpha} h^\alpha.$$

Furthermore, the rough-integral expansion gives

$$\begin{aligned} |R^Z_{s,t}| &= \left| \int_s^t Y \, \mathrm{d}\mathbf{X} - Y_s X_{s,t} \right| \\ &\lesssim |Y'_s| h^{2\alpha} + \|(Y, Y')\|_{X,2\alpha;[s,T]} h^{3\alpha} \\ &\lesssim s^{\eta\wedge 0-\alpha} h^{2\alpha}. \end{aligned}$$

When $h > s$, a dyadic decomposition reduces the estimates to the local case and sums a geometric series. These are exactly the two weighted bounds defining $\mathscr{D}_X^{2\alpha,\alpha+\eta\wedge 0}$. □

Remark. Singular controlled rough paths are the rough-path analogue of singular modelled distributions in regularity structures.

2.5 Uniqueness of the Gubinelli Derivative

Fix $\alpha \in (1/3, 1/2]$ and $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0, T]; V)$.

Definition 2.5.1 (True roughness). For $s \in [0, T)$, the path X is rough at time s if

$$\forall v^* \in V^* \setminus \{0\} : \qquad \limsup_{t \downarrow s} \frac{|v^*(X_{s,t})|}{|t-s|^{2\alpha}} = \infty.$$

It is truly rough if it is rough on a dense subset of $[0, T]$.

Proposition 2.5.2 (Uniqueness of the Gubinelli derivative). Let $(Y, Y') \in \mathscr{D}_X^{2\alpha}([0, T]; W)$ and suppose that X is rough at s. If

$$Y_{s,t} = O(|t-s|^{2\alpha}) \qquad (t \downarrow s),$$

then $Y'_s = 0$. Consequently, if X is truly rough and both (Y, Y') and $(Y, \widetilde{Y}')$ are controlled by X, then $Y' \equiv \widetilde{Y}'$.

Proof. The controlled expansion implies $Y'_s X_{s,t} = O(|t-s|^{2\alpha})$. For any $w^* \in W^*$, define $v^* = (Y'_s)^* w^* \in V^*$. Then
$$\frac{|v^*(X_{s,t})|}{|t-s|^{2\alpha}} = O(1).$$
Roughness forces $v^* = 0$. Since this holds for every w^*, $Y'_s = 0$. Apply this result to the controlled path $(0, Y' - \widetilde{Y}')$ at every rough time, then use continuity and density. □

Theorem 2.5.3 (Doob–Meyer principle for rough paths). Assume X is rough at s and $(Y, Y') \in \mathscr{D}_X^{2\alpha}$. Then
$$\int_s^t Y_u \, d\mathbf{X}_u = O(|t-s|^{2\alpha}) \quad (t \downarrow s) \quad \Longrightarrow \quad Y_s = 0.$$
If X is truly rough, $(\widetilde{Y}, \widetilde{Y}') \in \mathscr{D}_X^{2\alpha}$, and $Z, \widetilde{Z} \in \mathcal{C}([0,T]; W)$, then
$$\int_0^\cdot Y \, d\mathbf{X} + \int_0^\cdot Z_u \, du \equiv \int_0^\cdot \widetilde{Y} \, d\mathbf{X} + \int_0^\cdot \widetilde{Z}_u \, du$$
implies $(Y, Y') = (\widetilde{Y}, \widetilde{Y}')$ and $Z = \widetilde{Z}$.

Proof. The pair $(\int Y \, d\mathbf{X}, Y)$ is controlled by X, so the first claim is the preceding proposition applied to this controlled path. For the second, subtract the two identities. At every rough time s,
$$\int_s^t (Y_u - \widetilde{Y}_u) \, d\mathbf{X}_u = -\int_s^t (Z_u - \widetilde{Z}_u) \, du = O(|t-s|) = O(|t-s|^{2\alpha}),$$
because $2\alpha \le 1$. Hence $Y_s = \widetilde{Y}_s$ on a dense set and then everywhere by continuity. Uniqueness of the Gubinelli derivative gives $Y' = \widetilde{Y}'$. The remaining identity says that the integral of the continuous function $Z - \widetilde{Z}$ vanishes on every interval, so the fundamental theorem of calculus gives $Z = \widetilde{Z}$. □

Definition 2.5.4 (Quantitative Hölder roughness). A path $X \colon [0,T] \to V$ is θ-Hölder rough on scales at most ε_0 if there is $L_\theta(X) > 0$ such that for every $v^* \in V^*$, $s \in [0,T]$, and $\varepsilon \in (0, \varepsilon_0]$, there exists $t \in [0,T]$ with
$$|t-s| \le \varepsilon, \qquad |v^*(X_{s,t})| \ge L_\theta(X) \varepsilon^\theta |v^*|.$$
The largest possible constant is called the modulus of θ-Hölder roughness. For $v^* = 0$ the condition is automatic.

If $X \in \mathcal{C}^\alpha$ is θ-Hölder rough with $\theta < 2\alpha$, then it is truly rough.

Proposition 2.5.5 (Quantitative recovery of the derivative). If X is θ-Hölder rough and $(Y, Y') \in \mathscr{D}_X^{2\alpha}([0,T]; W)$, then for every $\varepsilon \in (0, \varepsilon_0]$,
$$L_\theta(X) \varepsilon^\theta \|Y'\|_\infty \le \operatorname{osc}(Y, \varepsilon) + \big(\|R^Y\|_{2\alpha} + \|Y'\|_\alpha \|X\|_\alpha \big) \varepsilon^{2\alpha},$$

where $\operatorname{osc}(Y,\varepsilon) := \sup_{|t-s|\le\varepsilon} |Y_{s,t}|$. If $\theta < 2\alpha$, this implies uniqueness of the Gubinelli derivative.

Proof. Our controlled remainder is defined on the ordered simplex. If the time provided by quantitative roughness satisfies $t \ge s$, then

$$|Y'_s X_{s,t}| \le \operatorname{osc}(Y,\varepsilon) + \|R^Y\|_{2\alpha}\varepsilon^{2\alpha}.$$

If $t < s$, write the controlled expansion from t to s and use $Y'_s X_{s,t} = Y'_t X_{s,t} + (Y'_s - Y'_t)X_{s,t}$. This gives the same bound with the additional term $\|Y'\|_\alpha \|X\|_\alpha \varepsilon^{2\alpha}$. For $w^* \in W^*$ with $|w^*| = 1$, apply quantitative roughness to $(Y'_s)^* w^* \in V^*$. Taking the supremum over w^*, then over s, proves the estimate. Apply it to $(0, Y' - \widetilde{Y}')$ and let $\varepsilon \downarrow 0$ when $\theta < 2\alpha$. □

Theorem 2.5.6 (Norris lemma for rough paths). Assume $\alpha \in (1/3, 1/2]$, let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T];V)$, and suppose that X is θ-Hölder rough with $\theta < 2\alpha$. Let $(Y,Y') \in \mathscr{D}_X^{2\alpha}([0,T];\mathcal{L}(V,W))$, $Z \in \mathcal{C}^\alpha([0,T];W)$, and

$$I_t := \int_0^t Y_u \,\mathrm{d}\mathbf{X}_u + \int_0^t Z_u \,\mathrm{d}u.$$

There exist $r, q > 0$ and $C = C(\alpha,\theta,T)$ such that, with

$$\mathcal{R} := 1 + L_\theta(X)^{-1} + \|\mathbf{X}\|_\alpha + |Y_0| + |Y'_0| + \|(Y,Y')\|_{X,2\alpha} + |Z_0| + \|Z\|_\alpha,$$

one has

$$\|Y\|_\infty + \|Z\|_\infty \le C\mathcal{R}^q \|I\|_\infty^r.$$

Proof sketch. Apply the preceding quantitative recovery estimate to the controlled path (I, Y). Its remainder consists of the 2α-Hölder rough-integral remainder plus the time integral of Z, which is Lipschitz. Optimizing the resulting estimate over the scale ε bounds $\|Y\|_\infty$ by a positive power of $\|I\|_\infty$, with polynomial dependence on $\mathcal{R}$. Interpolate between this supremum bound and the a priori controlled Hölder bounds to obtain a weaker Hölder norm of Y. Continuity of rough integration then controls $\|\int Y \,\mathrm{d}\mathbf{X}\|_\infty$. Since $\int Z \,\mathrm{d}t = I - \int Y \,\mathrm{d}\mathbf{X}$, interpolation between the supremum norm of the primitive and the α-Hölder norm of Z yields the corresponding bound for $\|Z\|_\infty$. Choosing the interpolation exponents in this order produces some $r, q > 0$ depending only on α and θ. □

Chapter 3 Rough Differential Equations

3.1 Young Differential Equations

As a warm-up for rough differential equations, we first treat equations driven by a path of Hölder regularity greater than one half.

Lemma 3.1.1 (Stability of composition on Hölder paths). Let $\alpha \in (0,1]$ and $f \in C_b^2$. There is a constant C, depending only on α, T, and $\|f\|_{C_b^2}$, such that

$$\|f(Y) - f(\widetilde{Y})\|_\alpha \le C(1 + \|Y\|_\alpha + \|\widetilde{Y}\|_\alpha)\big(|Y_0 - \widetilde{Y}_0| + \|Y - \widetilde{Y}\|_\alpha\big).$$

Proof. For every x, y,

$$f(x) - f(y) = \int_0^1 Df(y + r(x-y))(x-y)\,\mathrm{d}r.$$

Apply this identity at times s and t and subtract. Boundedness of Df and $D^2 f$ gives

$$|f(Y)_{s,t} - f(\widetilde{Y})_{s,t}| \lesssim |(Y - \widetilde{Y})_{s,t}| + (|Y_{s,t}| + |\widetilde{Y}_{s,t}|)\|Y - \widetilde{Y}\|_\infty.$$

Divide by $|t-s|^\alpha$ and use $\|Y - \widetilde{Y}\|_\infty \le |Y_0 - \widetilde{Y}_0| + T^\alpha \|Y - \widetilde{Y}\|_\alpha$. □

Proposition 3.1.2 (Well-posedness of Young differential equations). Let $\beta \in (1/2, 1]$, let $X \in \mathcal{C}^\beta([0,T];\mathbb{R}^d)$, let $f \in C_b^2(\mathbb{R}^m; \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$, and let $y \in \mathbb{R}^m$. There is a unique $Y \in \mathcal{C}^\beta([0,T];\mathbb{R}^m)$ satisfying

$$Y_t = y + \int_0^t f(Y_s)\,\mathrm{d}X_s.$$

Proof. Choose $\alpha \in (1/2, \beta)$ and, for $t \le T$, define

$$\mathcal{M}_t(Y)_r := y + \int_0^r f(Y_s)\,\mathrm{d}X_s, \qquad 0 \le r \le t.$$

On the complete set

$$\mathcal{B}_t := \{Y \in \mathcal{C}^\alpha([0,t];\mathbb{R}^m) : Y_0 = y,\ \|Y\|_{\alpha;[0,t]} \le 1\},$$

the Young estimate and the embedding $\|X\|_{\alpha;[0,t]} \le t^{\beta-\alpha}\|X\|_{\beta;[0,t]}$ give

$$\|\mathcal{M}_t(Y)\|_{\alpha;[0,t]} \le C_1 \|X\|_{\beta;[0,T]} t^{\beta-\alpha}.$$

Thus $\mathcal{M}_t$ maps $\mathcal{B}_t$ into itself when t is sufficiently small. For $Y, \widetilde{Y} \in \mathcal{B}_t$, the composition lemma and Young stability yield

$$\|\mathcal{M}_t(Y) - \mathcal{M}_t(\widetilde{Y})\|_{\alpha;[0,t]} \leq C_2 \|X\|_{\beta;[0,T]} t^{\beta-\alpha} \|Y - \widetilde{Y}\|_{\alpha;[0,t]}.$$

After reducing t further, the factor is at most $1/2$, so Banach's fixed-point theorem gives a unique local solution. The local time depends on the driver and f, but not on the current initial value because f and its derivatives are globally bounded. Iterating over finitely many intervals gives a unique global $\mathcal{C}^\alpha$ solution.

Finally, the equation and the Young estimate show directly that $Y_{s,t} = f(Y_s)X_{s,t} + O(|t-s|^{\alpha+\beta})$, so Y is actually β-Hölder. Uniqueness in $\mathcal{C}^\beta$ follows from uniqueness in $\mathcal{C}^\alpha$. □

Proposition 3.1.3 (Stability of Young differential equations). Let the preceding assumptions hold for two drivers $X, \widetilde{X}$ and two initial values $y, \widetilde{y}$. Suppose $\|X\|_\beta, \|\widetilde{X}\|_\beta \leq M$, and let $Y, \widetilde{Y}$ be the corresponding solutions. For every $\alpha \in (1/2, \beta)$,

$$\|Y - \widetilde{Y}\|_\alpha \leq C_M \big(|y - \widetilde{y}| + \|X - \widetilde{X}\|_\alpha\big),$$

where C_M depends on $\alpha, T, \|f\|_{C_b^2}$, and M.

Proof. On an interval $[s,t]$ short enough that the local contraction constants are uniformly below one, Young stability gives

$$\begin{aligned}\|Y - \widetilde{Y}\|_{\alpha;[s,t]} \leq C\Big(&|Y_s - \widetilde{Y}_s| + \|X - \widetilde{X}\|_{\alpha;[s,t]}\Big) \\ &+ \frac{1}{2}\|Y - \widetilde{Y}\|_{\alpha;[s,t]}.\end{aligned}$$

After absorption,

$$\|Y - \widetilde{Y}\|_{\alpha;[s,t]} \leq C\big(|Y_s - \widetilde{Y}_s| + \|X - \widetilde{X}\|_{\alpha;[s,t]}\big).$$

Choose a finite partition of $[0,T]$ by such intervals. At each right endpoint,

$$|Y_t - \widetilde{Y}_t| \leq |Y_s - \widetilde{Y}_s| + |t-s|^\alpha \|Y - \widetilde{Y}\|_{\alpha;[s,t]}.$$

A discrete Gronwall argument propagates the initial error and produces the global estimate. □

3.2 Estimates on Functions of Controlled Paths

We now move toward the fixed-point argument for rough differential equations.

Lemma 3.2.1 (Composition estimate). Let $\alpha \in (1/3, 1/2]$, $X \in \mathcal{C}^\alpha$, and $f \in C_b^2$. For every $(Y, Y') \in$

$\mathscr{D}_X^{2\alpha}$,

$$\big(f(Y), Df(Y)Y'\big) \in \mathscr{D}_X^{2\alpha}.$$

Moreover,

$$\|Df(Y)Y'\|_\alpha \le C\big(1 + |Y_0'| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha}\big)^2(1 + \|X\|_\alpha),$$
$$\|R^{f(Y)}\|_{2\alpha} \le C\big(1 + |Y_0'| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha}\big)^2(1 + \|X\|_\alpha)^2.$$

Proof. The product estimate gives

$$|Df(Y_t)Y_t' - Df(Y_s)Y_s'| \lesssim |Y_{s,t}'| + |Y_{s,t}|\,|Y_s'|.$$

Use the controlled-path estimate for $\|Y\|_\alpha$ to obtain the first bound. Taylor's theorem gives

$$R_{s,t}^{f(Y)} = f(Y_t) - f(Y_s) - Df(Y_s)Y_{s,t} + Df(Y_s)R_{s,t}^Y,$$

so

$$|R_{s,t}^{f(Y)}| \lesssim |Y_{s,t}|^2 + |R_{s,t}^Y|.$$

Divide by $|t-s|^{2\alpha}$ and use the same controlled-path bound. □

Lemma 3.2.2 (Stability of composition). Let $X, \widetilde{X} \in \mathcal{C}^\alpha$ and $(Y, Y') \in \mathscr{D}_X^{2\alpha}$, $(\widetilde{Y}, \widetilde{Y}') \in \mathscr{D}_{\widetilde{X}}^{2\alpha}$. Assume all driver and controlled-path norms appearing below are bounded by M, and let $f \in C_b^3$. Then

$$\|Df(Y)Y' - Df(\widetilde{Y})\widetilde{Y}'\|_\alpha + \|R^{f(Y)} - R^{f(\widetilde{Y})}\|_{2\alpha}$$
$$\le C_M\Big(|Y_0 - \widetilde{Y}_0| + |Y_0' - \widetilde{Y}_0'| + \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha} + \|X - \widetilde{X}\|_\alpha\Big).$$

Proof. For the derivatives, write

$$Df(Y)Y' - Df(\widetilde{Y})\widetilde{Y}' = Df(Y)(Y' - \widetilde{Y}') + (Df(Y) - Df(\widetilde{Y}))\widetilde{Y}'$$

and apply the Hölder product estimate and the composition stability lemma. The controlled decompositions give

$$\|Y - \widetilde{Y}\|_\alpha \le C_M(|Y_0' - \widetilde{Y}_0'| + \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha} + \|X - \widetilde{X}\|_\alpha).$$

For the remainders, use

$$R_{s,t}^{f(Y)} = \int_0^1 (1-r)D^2 f(Y_s + rY_{s,t})[Y_{s,t}, Y_{s,t}]\,\mathrm{d}r + Df(Y_s)R_{s,t}^Y,$$

subtract the analogous formula for $\widetilde{Y}$, and add and subtract one factor at a time. Boundedness of $D^2 f, D^3 f$ gives the stated bound. □

Lemma 3.2.3 (Rough integrals as controlled paths). Let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha$ and $f \in C_b^2$. For $(Y, Y') \in \mathscr{D}_X^{2\alpha}$, the pair

$$\left(\int_0^\cdot f(Y_u)\,\mathrm{d}\mathbf{X}_u,\ f(Y) \right) \in \mathscr{D}_X^{2\alpha}.$$

Furthermore,

$$\|f(Y)\|_\alpha \le C\Big((|Y_0'| + \|Y'\|_\alpha)\|X\|_\alpha + T^\alpha \|R^Y\|_{2\alpha} \Big),$$
$$\left\| R^{\int_0^\cdot f(Y_u)\,\mathrm{d}\mathbf{X}_u} \right\|_{2\alpha} \le C\big(1 + |Y_0'| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha}\big)^2 (1 + \|X\|_\alpha)^2 \|\mathbf{X}\|_\alpha.$$

Proof. The first estimate follows from Lipschitz continuity of f and the controlled expansion of Y. The integrand $f(Y)$ has Gubinelli derivative $Df(Y)Y'$. The rough-integral expansion therefore gives

$$\begin{aligned} R^I_{s,t} &:= \int_s^t f(Y_u)\,\mathrm{d}\mathbf{X}_u - f(Y_s)X_{s,t} \\ &= Df(Y_s)Y_s'\mathbb{X}_{s,t} + O(|t-s|^{3\alpha}). \end{aligned}$$

Use the rough-integral estimate together with the preceding composition bounds. □

Lemma 3.2.4 (Stability of rough-integral controlled paths). Let $\mathbf{X}, \widetilde{\mathbf{X}} \in \mathscr{C}^\alpha$ and let (Y, Y'), $(\widetilde{Y}, \widetilde{Y}')$ be controlled by their respective first levels. Under a common bound M and for $f \in C_b^3$,

$$\begin{aligned} \|f(Y) - f(\widetilde{Y})\|_\alpha \le C_M\Big(&|Y_0 - \widetilde{Y}_0| + |Y_0' - \widetilde{Y}_0'| + \|Y' - \widetilde{Y}'\|_\alpha \\ &+ \|R^Y - R^{\widetilde{Y}}\|_{2\alpha} + \|X - \widetilde{X}\|_\alpha \Big), \end{aligned}$$

and, writing $I = \int f(Y)\,\mathrm{d}\mathbf{X}$ and $\widetilde{I} = \int f(\widetilde{Y})\,\mathrm{d}\widetilde{\mathbf{X}}$,

$$\begin{aligned} \|R^I - R^{\widetilde{I}}\|_{2\alpha} \le C_M\Big[&\big(|Y_0 - \widetilde{Y}_0| + |Y_0' - \widetilde{Y}_0'| + \|Y' - \widetilde{Y}'\|_\alpha \\ &+ \|R^Y - R^{\widetilde{Y}}\|_{2\alpha} + \|X - \widetilde{X}\|_\alpha\big)\|\mathbf{X}\|_\alpha + \|\mathbf{X}; \widetilde{\mathbf{X}}\|_\alpha \Big]. \end{aligned}$$

Proof. The first estimate is the Hölder composition estimate combined with the controlled-path stability bound. For the second, apply the stability theorem for rough-integral remainders to the controlled integrands $(f(Y), Df(Y)Y')$ and $(f(\widetilde{Y}), Df(\widetilde{Y})\widetilde{Y}')$, then invoke the preceding composition-stability lemma. □

3.3 Well-posedness of Rough Differential Equations

Consider

$$\mathrm{d}Y_t = f(Y_t)\,\mathrm{d}\mathbf{X}_t.$$

Theorem 3.3.1 (Well-posedness of RDEs). Let $\beta \in (1/3, 1/2]$, $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\beta([0,T]; \mathbb{R}^d)$, $f \in C_b^3(\mathbb{R}^m; \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$, and $y \in \mathbb{R}^m$. There is a unique $(Y, Y') \in \mathscr{D}_X^{2\beta}([0,T]; \mathbb{R}^m)$ such that $Y' = f(Y)$ and

$$Y_t = y + \int_0^t f(Y_s)\, \mathrm{d}\mathbf{X}_s.$$

Proof. Choose

$$\max\{1/3, 2\beta/3\} < \alpha < \beta.$$

For $t \le T$, define

$$\mathcal{M}_t(Y, Y') := \left(y + \int_0^\cdot f(Y_s)\, \mathrm{d}\mathbf{X}_s,\ f(Y) \right)$$

on $[0,t]$. Consider the closed ball in $\mathscr{D}_X^{2\alpha}$ with fixed initial data $Y_0 = y$, $Y_0' = f(y)$ and controlled seminorm at most one. The rough-integral and composition estimates show that

$$\|\mathcal{M}_t(Y, Y')\|_{X, 2\alpha; [0,t]} \le C_1 \Big(\|X\|_{\beta;[0,t]} t^{\beta - \alpha} + \|\mathbb{X}\|_{2\beta;[0,t]} t^{2(\beta-\alpha)} + t^\alpha \Big).$$

Thus the ball is invariant for small t. The corresponding stability estimates give, for two elements with the same initial data,

$$\begin{aligned}
&\|\mathcal{M}_t(Y, Y') - \mathcal{M}_t(\widetilde{Y}, \widetilde{Y}')\|_{X, 2\alpha; [0,t]} \\
&\quad \le C_2 \Big(\|X\|_{\beta;[0,t]} t^{\beta - \alpha} + \|\mathbb{X}\|_{2\beta;[0,t]} t^{2(\beta-\alpha)} + t^\alpha \Big) \|(Y, Y') - (\widetilde{Y}, \widetilde{Y}')\|_{X, 2\alpha; [0,t]}.
\end{aligned}$$

After reducing t, this factor is at most $1/2$. Banach's fixed-point theorem therefore yields a unique local solution. The local time is uniform in the current initial value, so finitely many intervals cover $[0,T]$.

It remains to recover the driver regularity. The fixed-point identity gives

$$Y_{s,t} = f(Y_s) X_{s,t} + Df(Y_s) f(Y_s) \mathbb{X}_{s,t} + O(|t-s|^{3\alpha}).$$

Because $3\alpha > 2\beta$, this implies $Y \in \mathcal{C}^\beta$ and $R^Y \in \mathcal{C}_2^{2\beta}$. Since f is Lipschitz, $Y' = f(Y) \in \mathcal{C}^\beta$. Hence the solution belongs to $\mathscr{D}_X^{2\beta}$, and uniqueness follows from the local contraction. $\square$

The map sending $(y, \mathbf{X})$ to the RDE solution is called the Itô–Lyons map.

Theorem 3.3.2 (Local Lipschitz continuity of the Itô–Lyons map). Let $\beta \in (1/3, 1/2]$ and $f \in C_b^3$. Let $\mathbf{X}, \widetilde{\mathbf{X}} \in \mathscr{C}^\beta$ satisfy

$$\|\mathbf{X}\|_\beta,\ \|\widetilde{\mathbf{X}}\|_\beta \le M,$$

and let (Y, Y'), $(\widetilde{Y}, \widetilde{Y}')$ be the RDE solutions with initial conditions $y, \tilde{y}$. For every $\alpha \in (1/3, \beta)$ there is C_M such that

$$\|Y - \widetilde{Y}\|_\alpha + \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha} \le C_M \big(|y - \tilde{y}| + \|\mathbf{X}; \widetilde{\mathbf{X}}\|_\alpha \big).$$

Proof. The a priori estimates from the existence proof give a common bound for both solutions on sufficiently short intervals, with the interval length depending only on M, α, β, and f. On such an interval $[s, t]$, composition and rough-integral stability yield

$$\begin{aligned}
&\|Y' - \widetilde{Y}'\|_{\alpha;[s,t]} + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha;[s,t]} \\
&\quad \leq C\Big(|Y_s - \widetilde{Y}_s| + \|\mathbf{X}; \widetilde{\mathbf{X}}\|_{\alpha;[s,t]}\Big) + \frac{1}{2}\Big(\|Y' - \widetilde{Y}'\|_{\alpha;[s,t]} + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha;[s,t]}\Big).
\end{aligned}$$

After absorption, controlled-path stability also gives

$$\|Y - \widetilde{Y}\|_{\alpha;[s,t]} \leq C\Big(|Y_s - \widetilde{Y}_s| + \|\mathbf{X}; \widetilde{\mathbf{X}}\|_{\alpha;[s,t]}\Big).$$

The endpoint error satisfies

$$|Y_t - \widetilde{Y}_t| \leq |Y_s - \widetilde{Y}_s| + |t - s|^\alpha \|Y - \widetilde{Y}\|_{\alpha;[s,t]}.$$

Partition $[0, T]$ into finitely many such intervals and use discrete Gronwall. This propagates the current endpoint error correctly and yields the global bound. $\square$

Chapter 4 Connections with Stochastic Calculus

4.1 Brownian Motion as a Rough Path

We begin with a two-level version of Kolmogorov's continuity criterion. The first level is a one-parameter process, whereas the second level is a measurable two-parameter process on the simplex.

Theorem 4.1.1 (Kolmogorov criterion for rough paths). Let

$$X\colon \Omega \times [0,T] \to \mathbb{R}^d, \qquad \mathbb{X}\colon \Omega \times \Delta_{[0,T]} \to \mathbb{R}^d \otimes \mathbb{R}^d$$

be measurable, and write $X_{s,t} = X_t - X_s$. Assume that on one event of probability one,

$$\mathbb{X}_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t} \qquad (0 \le s \le u \le t \le T).$$

Let $q \geq 2$ and $\beta > 1/q$. Suppose that a constant C satisfies

$$\|X_{s,t}\|_{L^q} \le C|t-s|^{\beta}, \qquad \|\mathbb{X}_{s,t}\|_{L^{q/2}} \le C|t-s|^{2\beta}$$

for every $(s,t) \in \Delta_{[0,T]}$. Then, for every $\alpha \in (0, \beta - 1/q)$, there is a modification $(\widetilde{X}, \widetilde{\mathbb{X}})$ and random variables $K_\alpha \in L^q$ and $\mathbb{K}_\alpha \in L^{q/2}$ such that

$$|\widetilde{X}_{s,t}| \le K_\alpha |t-s|^{\alpha}, \qquad |\widetilde{\mathbb{X}}_{s,t}| \le \mathbb{K}_\alpha |t-s|^{2\alpha}$$

for all $0 \le s \le t \le T$. The modification satisfies Chen's relation on one event of probability one. In particular, if $\beta - 1/q > 1/3$, then $(\widetilde{X}, \widetilde{\mathbb{X}}) \in \mathscr{C}^\alpha$ almost surely for every $\alpha \in (1/3, \beta - 1/q)$.

Proof. A linear change of time reduces the proof to $T = 1$. Let

$$D_n := \{k2^{-n} : k = 0, \dots, 2^n\}$$

and define the largest increments over dyadic intervals of level n by

$$K_n := \max_{0 \le k < 2^n} |X_{k2^{-n},(k+1)2^{-n}}|,$$
$$\mathbb{K}_n := \max_{0 \le k < 2^n} |\mathbb{X}_{k2^{-n},(k+1)2^{-n}}|.$$

The moment assumptions imply

$$\mathbb{E}[K_n^q] \le C^q 2^{-n(\beta q - 1)}, \qquad \mathbb{E}[\mathbb{K}_n^{q/2}] \le C^{q/2} 2^{-n(\beta q - 1)}.$$

Consequently,

$$\|K_n\|_{L^q} \le C2^{-n(\beta-1/q)}, \qquad \|\mathbb{K}_n\|_{L^{q/2}} \le C2^{-2n(\beta-1/q)}.$$

Fix dyadic $s < t$ and choose m so that $2^{-(m+1)} < t - s \le 2^{-m}$. The interval $[s,t]$ admits a disjoint dyadic decomposition containing at most two intervals at every level $n \ge m+1$. If

$$s = u_0 < u_1 < \cdots < u_N = t$$

is the resulting partition, additivity of the first level gives

$$|X_{s,t}| \le \sum_{i=0}^{N-1} |X_{u_i,u_{i+1}}| \le 2 \sum_{n=m+1}^{\infty} K_n.$$

Iterating Chen's relation gives

$$\mathbb{X}_{s,t} = \sum_{i=0}^{N-1} \mathbb{X}_{u_i,u_{i+1}} + \sum_{i=0}^{N-1} X_{s,u_i} \otimes X_{u_i,u_{i+1}}.$$

Hence

$$|\mathbb{X}_{s,t}| \le 2 \sum_{n=m+1}^{\infty} \mathbb{K}_n + \left(2 \sum_{n=m+1}^{\infty} K_n\right)^2.$$

Define

$$K_\alpha := 2 \sum_{n=0}^{\infty} 2^{n\alpha} K_n,$$

$$\mathbb{K}_\alpha := 2 \sum_{n=0}^{\infty} 2^{2n\alpha} \mathbb{K}_n + K_\alpha^2.$$

Since $\alpha < \beta - 1/q$, Minkowski's inequality and the preceding moment bounds show that $K_\alpha \in L^q$ and $\mathbb{K}_\alpha \in L^{q/2}$. Moreover, for every pair of dyadic times,

$$|X_{s,t}| \le K_\alpha |t-s|^\alpha, \qquad |\mathbb{X}_{s,t}| \le \mathbb{K}_\alpha |t-s|^{2\alpha}.$$

These bounds hold simultaneously outside one null set.

On that event, X has a unique continuous extension from the dyadic times; denote it by $\widetilde{X}$. To extend the second level, first put $A_t := \mathbb{X}_{0,t}$ at dyadic times. Chen's relation and the bounds above give, for dyadic $s < t$,

$$|A_t - A_s| \le |\mathbb{X}_{s,t}| + |X_{0,s}|\,|X_{s,t}|,$$

so A also has a unique continuous extension. Define

$$\widetilde{\mathbb{X}}_{s,t} := A_t - A_s - \widetilde{X}_{0,s} \otimes \widetilde{X}_{s,t}.$$

This definition agrees with $\mathbb{X}$ at dyadic pairs and automatically satisfies Chen's relation. Approximating arbitrary $s < t$ by dyadic pairs and passing to the limit gives the asserted 2α-Hölder bound.

It remains to check the modification property. For each fixed t, the moment estimate gives $X_{t_n} \to X_t$ in L^q whenever dyadic $t_n \to t$; therefore $\widetilde{X}_t = X_t$ almost surely. For a fixed $s < t$, choose dyadic sequences with $s_n \uparrow s$ and $t_n \downarrow t$, so that $s_n \leq s < t \leq t_n$. Chen's relation yields

$$\mathbb{X}_{s_n,t_n} - \mathbb{X}_{s,t} = \mathbb{X}_{s_n,s} + \mathbb{X}_{t,t_n} + X_{s_n,s} \otimes X_{s,t_n} + X_{s,t} \otimes X_{t,t_n}.$$

Hölder's inequality and the assumed moment estimates show that the right-hand side tends to zero in $L^{q/2}$. Thus $\widetilde{\mathbb{X}}_{s,t} = \mathbb{X}_{s,t}$ almost surely for every fixed pair (s,t). □

Remark. After choosing the continuous modification, it is customary to suppress the tildes and continue to write $(X, \mathbb{X})$.

Let $B = (B^1, \dots, B^d)$ be standard Brownian motion. Its Itô second level is defined componentwise by

$$\mathbb{B}^{\text{Itô}}_{s,t} := \int_s^t B_{s,r} \otimes \mathrm{d}B_r, \qquad B_{s,r} := B_r - B_s.$$

Splitting the stochastic integral at an intermediate time and using $B_{s,r} = B_{s,u} + B_{u,r}$ shows that $(B, \mathbb{B}^{\text{Itô}})$ satisfies Chen's relation.

Proposition 4.1.2 (Itô Brownian rough path). For every $\alpha \in (1/3, 1/2)$, after choosing a continuous modification,

$$\mathbf{B}^{\text{Itô}} := (B, \mathbb{B}^{\text{Itô}}) \in \mathscr{C}^\alpha([0,T]; \mathbb{R}^d)$$

almost surely.

Proof. For every $q \geq 4$, Brownian scaling gives

$$\|B_{s,t}\|_{L^q} \leq C_q |t-s|^{1/2}.$$

The Burkholder–Davis–Gundy inequality and Minkowski's integral inequality give

$$\begin{aligned}\|\mathbb{B}^{\text{Itô}}_{s,t}\|_{L^{q/2}} &\leq C_q \left\| \left(\int_s^t |B_{s,r}|^2 \, \mathrm{d}r \right)^{1/2} \right\|_{L^{q/2}} \\ &\leq C_q \left(\int_s^t \|B_{s,r}\|^2_{L^{q/2}} \, \mathrm{d}r \right)^{1/2} \leq C_q |t-s|.\end{aligned}$$

Fix $\alpha < 1/2$ and choose one finite q large enough that $\alpha < 1/2 - 1/q$. The preceding theorem, with $\beta = 1/2$, then gives the claim. No limiting argument in q is needed. □

The Stratonovich second level is

$$\mathbb{B}^{\text{Strat}}_{s,t} := \int_s^t B_{s,r} \otimes \circ \mathrm{d}B_r.$$

Componentwise Itô–Stratonovich conversion gives

$$\mathbb{B}^{\text{Strat}}_{s,t} = \mathbb{B}^{\text{Itô}}_{s,t} + \frac{1}{2}(t-s) I_d.$$

In particular,

$$\mathrm{Sym}(\mathbb{B}^{\mathrm{Strat}}_{s,t}) = \frac{1}{2} B_{s,t} \otimes B_{s,t}.$$

Proposition 4.1.3 (Stratonovich Brownian rough path). For every $\alpha \in (1/3, 1/2)$,

$$\mathbf{B}^{\mathrm{Strat}} := (B, \mathbb{B}^{\mathrm{Strat}}) \in \mathscr{C}^{0,\alpha}_g([0,T]; \mathbb{R}^d)$$

almost surely.

Proof. The deterministic correction $(s,t) \mapsto \frac{1}{2}(t-s) I_d$ is 1-Hölder, so the preceding proposition gives $\mathbf{B}^{\mathrm{Strat}} \in \mathscr{C}^{\beta}$ almost surely for every $\beta < 1/2$. The symmetry identity shows that it is weakly geometric, hence $\mathbf{B}^{\mathrm{Strat}} \in \mathscr{C}^{\beta}_g$. Given $1/3 < \alpha < \beta < 1/2$, the inclusion $\mathscr{C}^{\beta}_g \subset \mathscr{C}^{0,\alpha}_g$ proves the claim. Equivalently, one may use the canonical lifts of piecewise linear Brownian approximations, which converge to the Stratonovich lift in every lower Hölder rough-path topology. □

4.2 Consistency with Stochastic Calculus

All stochastic integrals below are taken componentwise. Let $Y_t \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)$ and

$$Y'_t \in \mathcal{L}\big(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)\big).$$

We regard Y'_t as a bilinear map and use the contraction

$$\mathrm{Ctr}(Y'_t) := \sum_{i=1}^{d} Y'_t(e_i) e_i \in \mathbb{R}^m,$$

where $(e_i)_{i=1}^d$ is the standard basis of $\mathbb{R}^d$.

Proposition 4.2.1 (Consistency of the Itô integral). Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, and let B be an $(\mathcal{F}_t)$-Brownian motion. Fix $\alpha \in (1/3, 1/2)$ and let $\mathbf{B}^{\text{Itô}}$ be its Itô rough-path lift. Suppose that (Y, Y') is continuous and adapted and that, almost surely,

$$(Y, Y') \in \mathscr{D}^{2\alpha}_B([0,T]; \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)).$$

Then Y is locally square integrable and, for every $t \in [0,T]$,

$$\int_0^t Y_u \, \mathrm{d}\mathbf{B}^{\text{Itô}}_u = \int_0^t Y_u \, \mathrm{d}B_u$$

almost surely. The equality holds simultaneously for all t after choosing continuous versions.

Proof. Continuity implies $\int_0^T |Y_u|^2 \, \mathrm{d}u < \infty$ almost surely, so the adapted process Y is locally square integrable. Let (π^n) be any sequence of deterministic partitions of $[0,t]$ whose mesh tends to zero. The rough Riemann sums are

$$S_n := \sum_{[u,v]\in\pi^n} \left(Y_u B_{u,v} + Y'_u \mathbb{B}^{\text{Itô}}_{u,v}\right),$$

and converge almost surely to the rough integral. The first sums $\sum Y_u B_{u,v}$ converge in probability to the Itô integral.

It remains to show that

$$R_n := \sum_{[u,v]\in\pi^n} Y'_u \mathbb{B}^{\text{Itô}}_{u,v} \longrightarrow 0$$

in probability. First suppose $|Y'_u| \le M$. The random variables $Y'_u \mathbb{B}^{\text{Itô}}_{u,v}$ are martingale differences over disjoint partition intervals. Orthogonality and the estimate $\mathbb{E}|\mathbb{B}^{\text{Itô}}_{u,v}|^2 \le C|v-u|^2$ yield

$$\mathbb{E}|R_n|^2 = \sum_{[u,v]\in\pi^n} \mathbb{E}|Y'_u \mathbb{B}^{\text{Itô}}_{u,v}|^2 \le CM^2T|\pi^n|.$$

Thus $R_n \to 0$ in L^2.

For general Y', let θ_M be a bounded radial truncation equal to the identity on the ball of radius M. The same estimate applies to $\theta_M(Y')$. Since $\sup_{u\le T}|Y'_u| < \infty$ almost surely,

$$\begin{aligned}\mathbb{P}(|R_n| > \varepsilon) \le \mathbb{P}&\left(\sup_{u\le T}|Y'_u| > M\right) \\ &+ \mathbb{P}\left(\left|\sum_{[u,v]\in\pi^n} \theta_M(Y'_u)\mathbb{B}^{\text{Itô}}_{u,v}\right| > \varepsilon\right).\end{aligned}$$

Letting first $n\to\infty$ and then $M\to\infty$ proves convergence in probability. Hence S_n converges in probability both to the rough integral and to the Itô integral; uniqueness of limits in probability gives equality almost surely. Applying the argument at rational times and using continuity of both integral processes gives simultaneous equality for every t. □

For a continuous adapted matrix-valued process Y, write

$$[Y,B]^{\text{ctr}}_t := \sum_{i=1}^{d} [\, t \mapsto Y_t e_i, B^i\,]_t$$

whenever the componentwise quadratic covariations exist. The corresponding Stratonovich integral is

$$\int_0^t Y_u \circ \mathrm{d}B_u := \int_0^t Y_u \,\mathrm{d}B_u + \frac{1}{2}[Y,B]^{\text{ctr}}_t.$$

Proposition 4.2.2 (Consistency of the Stratonovich integral). Under the assumptions of the preceding proposition, replace the Itô lift by $\mathbf{B}^{\text{Strat}}$. Then the contracted quadratic covariation exists and satisfies

$$[Y,B]^{\text{ctr}}_t = \int_0^t \mathrm{Ctr}(Y'_u)\,\mathrm{d}u.$$

Consequently,
$$\int_0^t Y_u \,\mathrm{d}\mathbf{B}_u^{\mathrm{Strat}} = \int_0^t Y_u \circ \mathrm{d}B_u$$
almost surely, simultaneously for every $t \in [0,T]$.

Proof. For a partition interval $[u,v]$, controlledness gives
$$Y_{u,v} = Y'_u B_{u,v} + R^Y_{u,v}.$$
Contracting with $B_{u,v}$ and using
$$B_{u,v} \otimes B_{u,v} = 2\,\mathrm{Sym}(\mathbb{B}^{\mathrm{Itô}}_{u,v}) + (v-u) I_d$$
gives
$$Y_{u,v} B_{u,v} = 2Y'_u\,\mathrm{Sym}(\mathbb{B}^{\mathrm{Itô}}_{u,v}) + \mathrm{Ctr}(Y'_u)(v-u) + R^Y_{u,v} B_{u,v}.$$
The martingale-difference argument from the Itô proposition, applied also to the transposed second level, shows that the sum of the first terms tends to zero in probability. The middle terms are ordinary Riemann sums and converge almost surely to $\int_0^t \mathrm{Ctr}(Y'_u)\,\mathrm{d}u$. Finally,
$$\left| \sum_{[u,v]\in\pi} R^Y_{u,v} B_{u,v} \right| \le \|R^Y\|_{2\alpha} \|B\|_\alpha \sum_{[u,v]\in\pi} |v-u|^{3\alpha} \le T \|R^Y\|_{2\alpha} \|B\|_\alpha |\pi|^{3\alpha-1},$$
which tends to zero because $3\alpha > 1$. This identifies the contracted quadratic covariation.

At the level of rough Riemann sums,
$$\mathbb{B}^{\mathrm{Strat}}_{u,v} = \mathbb{B}^{\mathrm{Itô}}_{u,v} + \frac{1}{2}(v-u) I_d$$
implies
$$\int_0^t Y_u \,\mathrm{d}\mathbf{B}_u^{\mathrm{Strat}} = \int_0^t Y_u \,\mathrm{d}\mathbf{B}_u^{\mathrm{Itô}} + \frac{1}{2}\int_0^t \mathrm{Ctr}(Y'_u)\,\mathrm{d}u.$$
The Itô consistency result and the formula just proved turn the right-hand side into the classical Stratonovich integral. □

The consistency of differential equations is now a consequence of the integral identities, but adaptation must be justified by the causal nature of the Itô–Lyons solution map rather than by continuity alone.

Proposition 4.2.3 (Consistency of RDEs and SDEs). Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ satisfy the usual conditions. Let B be a d-dimensional $(\mathcal{F}_t)$-Brownian motion, let $y \in L^2(\Omega; \mathbb{R}^m)$ be $\mathcal{F}_0$-measurable, and let
$$f \in C^3_b(\mathbb{R}^m; \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)).$$
Fix $\alpha \in (1/3, 1/2)$.

(a). Let (Y, Y') solve

$$dY_t = f(Y_t)\, d\mathbf{B}_t^{\text{Itô}}, \qquad Y_0 = y.$$

Then Y is the unique strong solution of the Itô SDE

$$dY_t = f(Y_t)\, dB_t, \qquad Y_0 = y.$$

(b). Let (Y, Y') solve

$$dY_t = f(Y_t)\, d\mathbf{B}_t^{\text{Strat}}, \qquad Y_0 = y.$$

Then Y is the unique strong solution of the Stratonovich SDE

$$dY_t = f(Y_t) \circ dB_t, \qquad Y_0 = y.$$

Proof. For each t, the restricted enhanced path $\mathbf{B}|_{[0,t]}$ is $\mathcal{F}_t$-measurable. By uniqueness of RDE solutions, the restriction $Y|_{[0,t]}$ is obtained by applying the deterministic Itô–Lyons map to y and this restricted driver; in particular, the construction is nonanticipative. Continuity, hence Borel measurability, of that map shows that Y_t is $\mathcal{F}_t$-measurable. Thus Y and $Y' = f(Y)$ are adapted. In the Itô case, the preceding consistency proposition applied on every interval $[0, t]$ gives

$$Y_t = y + \int_0^t f(Y_u)\, d\mathbf{B}_u^{\text{Itô}} = y + \int_0^t f(Y_u)\, dB_u.$$

Hence Y solves the Itô SDE. Because $f \in C_b^3$, it is globally Lipschitz, so pathwise uniqueness holds; the adapted solution constructed from B and y is therefore the unique strong solution. The Stratonovich consistency proposition gives the corresponding integral equation in the second case. Equivalently, conversion to Itô form produces bounded globally Lipschitz drift and diffusion coefficients, so the same strong existence and uniqueness conclusion applies. $\square$